Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Tao, Sarah Ph.D.		
Project Title:	Microdevices for Controlled, Targeted Oral Drug Delivery		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NON-PROFIT	**Phone:**	617-258-4855
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City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	09/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Medications are commonly administered for both routine medical care and countering adverse effects of space flight. Low bioavailability in microgravity has increased the necessity to develop novel therapeutic modalities for effective remote site treatment. In response, the proposed research outlines the development of a device which will provide a self-administered, non-invasive system for efficient drug delivery. Microdevices for controlled, targeted oral drug delivery will exploit the ability of microelectromechanical systems (MEMS) technology and surface chemistry to precisely tailor features and bioadhesive characteristics of the delivery vehicle. This system may allow controlled, passive release of medication, benefiting patients in space and on Earth.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	The remainder of the GSRP 2004 funding and renewal funding for GSRP 2005 will be used to continue and build upon the research outlined in the original proposal. The fabrication and modification of the microdevices has been proven feasible and designs have been constructed to incorporate mechanical properties which may allow the devices to adhere chemically as well as physically. Novel microdevice features will be continuously be developed. Additional in vitro experimentation will be conducted to study the effect induced by construct bound lectin as well as free lectin. Tight junction proteins ZO-1, and occludin will be examined using immunochemistry to observe any correlation with significant increase/decrease in TEER. In addition, transmission electron microscopy is being used to determine the effect of the bioadhesive device/sphere on intracellular structures and microvillar ultrastructure. Currently, a dynamic flow chamber to test the bioadhesion of themicrodevices is being set up. This system will allow careful control and calibration of flow rates and particle administration, plus the ability to image dynamic particle adhesion to cells in vitro. Experiments to study the behavior of bioadhesive delivery devices under dynamic conditions, mimicking the in vivo environment will be developed. In addition, release from the microdevices will be characterized using BSA as a model drug in situ. Once varied release profiles have been achieved, bioavailability will be examined both in vitro and in vivo.
Bibliography Type:	Description: (Last Updated: 01/03/2008)
Articles in Peer-reviewed Journals	Tao, S.L.; Desai, T.A. "Micromachined Polymeric Devices for Applications in Targeted Drug Delivery" JALA , Jun-2004
Articles in Peer-reviewed Journals	Tao, S.L.; Desai, T.A. "Microfabrication of multilayer, asymmetric, polymeric devices for drug delivery" Advanced Materials , Jul-2005
Articles in Peer-reviewed Journals	Tao SL, Desai TA "Micromachined Devices: The impact of controlled delivery from cell targeting to bioavailability" J Control Release. 2005 Dec; 109(1-3): 127-138. , Dec-2005